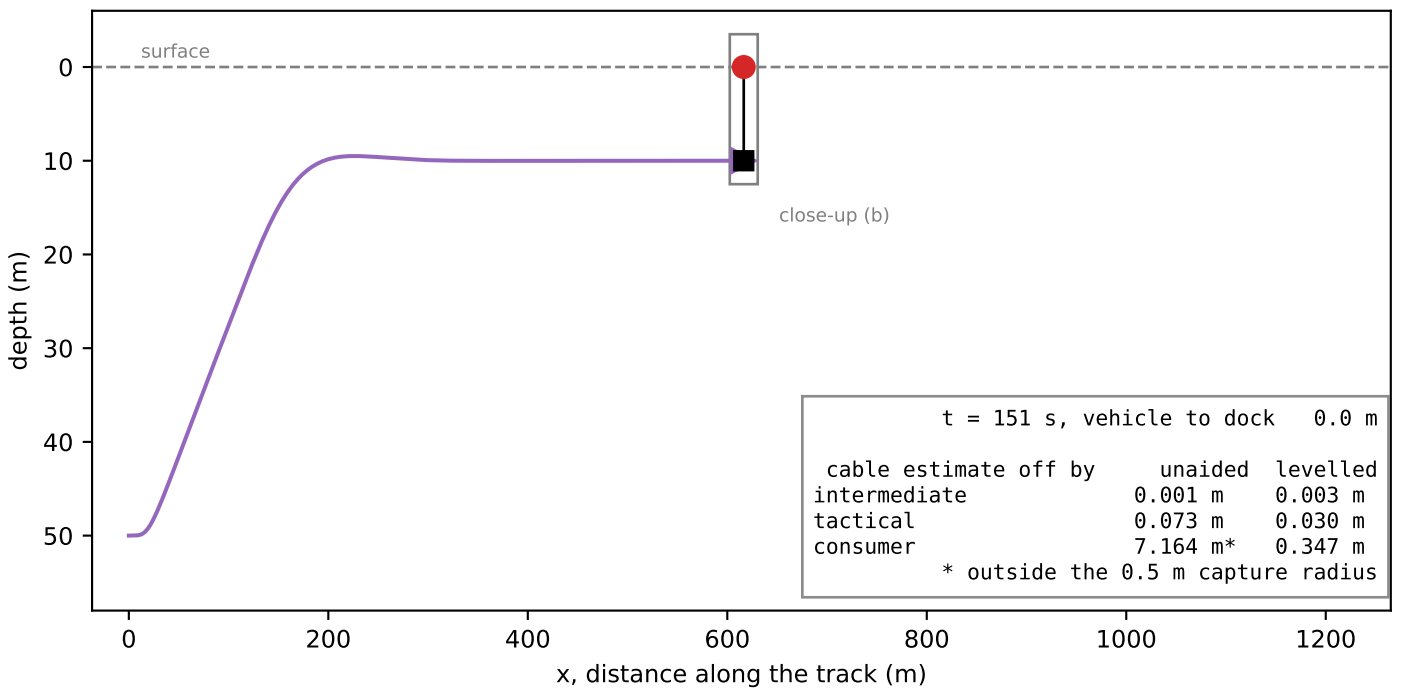
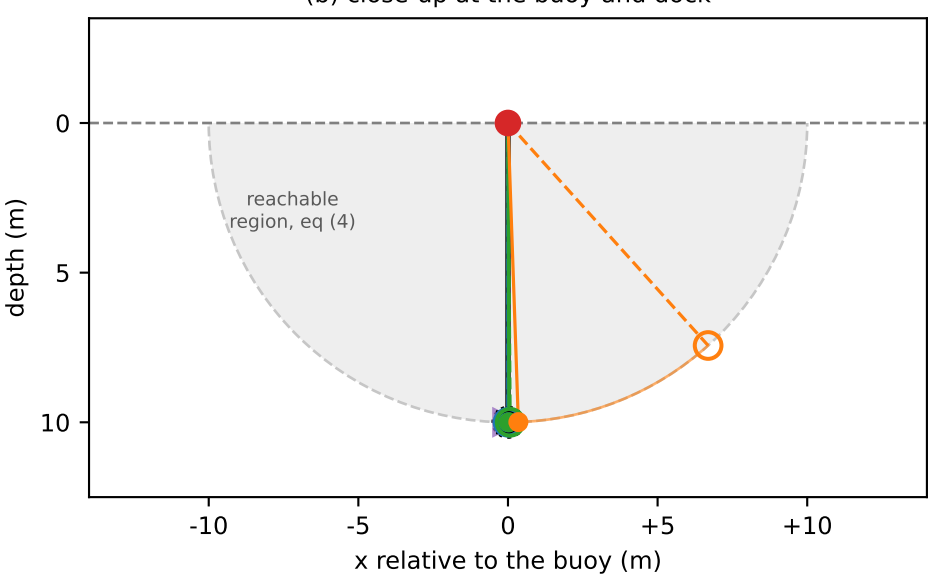
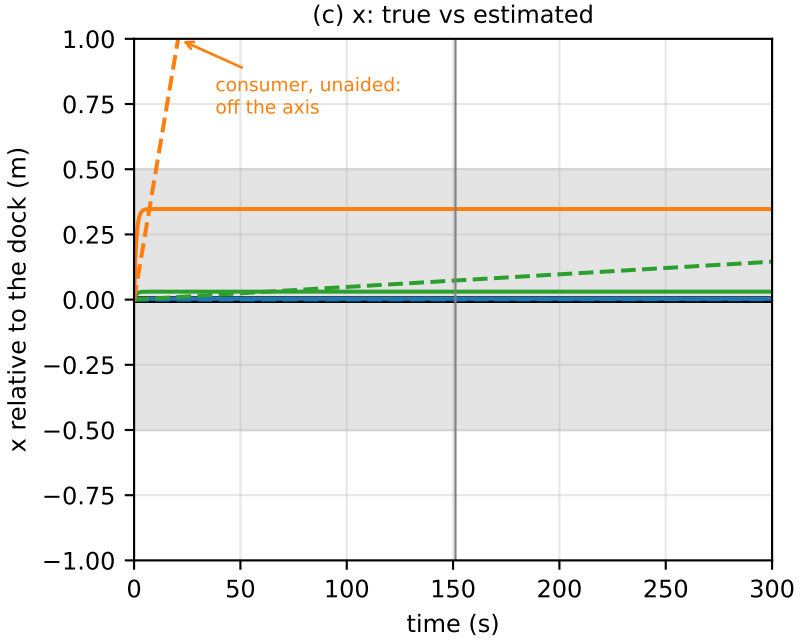
Dock position estimated through the cable, eq (3), from the attitude its own navigation unit gives,
for three IMU grades - black is the truth, which never moves

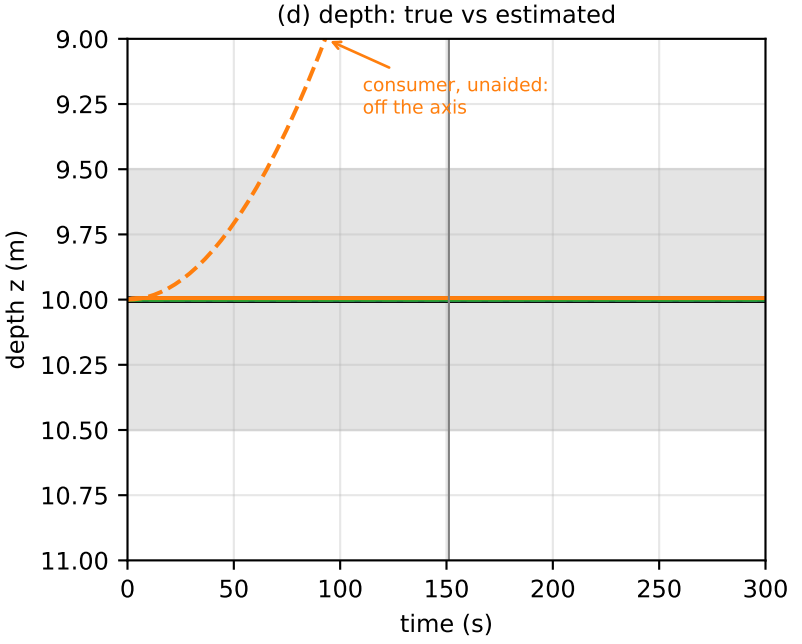
(a) the scene: the vehicle holds the dock's depth and passes it at t = 151 s
(it has no sensing of the dock yet; it gives the scale)

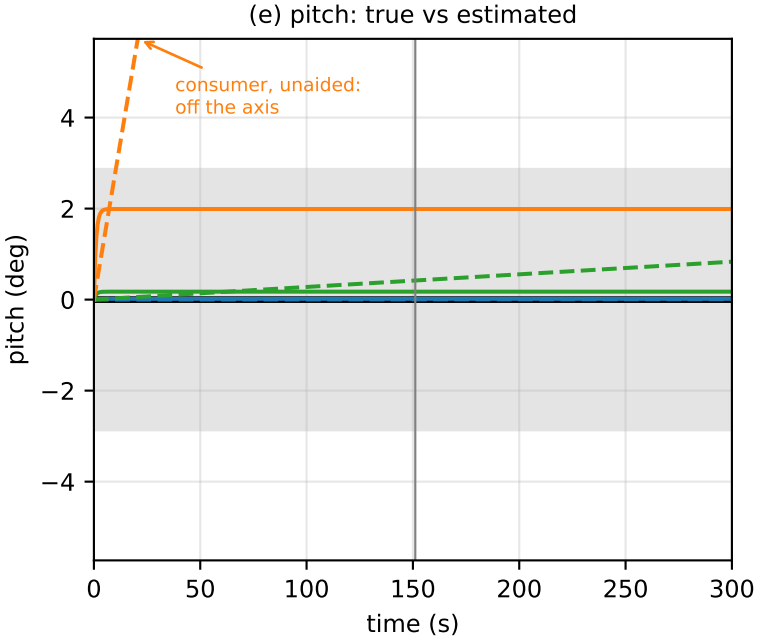


(b) close-up at the buoy and dock









buoy: source, position known
true dock and cable (never moves)
vehicle
reachable region, eq (4): $\|p_D - p_S\| \le L$
capture radius 0.5 m (in pitch: the error that reaches it)
dock estimate, eq (3): intermediate, 0.1 deg/hr, 0.3 mg
dock estimate, eq (3): tactical, 10 deg/hr, 3 mg
dock estimate, eq (3): consumer, 1000 deg/hr, 30 mg
unaided: gyro alone
levelled: complementary filter, τ = 1 s

IMU grades from Groves' table of typical biases (consumer: one decade past its lower bounds). The dock's INS starts from its true attitude (alignment assumed perfect).
Its gyro bias is about the pitch axis and its accelerometer bias along x, so y, roll and yaw stay exact. The vehicle and buoy have ideal IMUs.